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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtproto kernel_r
        valid_lft 179068sec preferred_lft 83386sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:                0.0
Usage of /:                  5.1% of 48.91GB
Memory usage:               5%
Swap usage:                 0%
Processes:                  139
Users logged in:            1
IPv4 address for enp0s3:    192.168.1.14
IPv6 address for enp0s3:    2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSIT 3B

Date Submitted: September 2, 2026

Instructor: Sir Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☒ ~~Create and configure a Virtual Machine with Ubuntu Server~~
- ☒ ~~Configure Bridged Network Adapter for network access~~
- ☒ ~~Determine and verify the server's IP address~~
- ☒ ~~Install OpenSSH Server and enable the service~~
- ☒ ~~Establish successful SSH connection from host machine~~
- ☒ ~~Execute verification commands to confirm remote access~~

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu Server 26
Virtualization Software (VirtualBox/VMware)	Oracle VM Virtualbox
Ubuntu Server Version	Ubuntu Server 26.04 LTS
VM Name	SaulogServer
RAM Allocated	4096
Storage Allocated	100 GB
Network Mode	Bridged Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	<u>4096</u> MB
Hard Disk	<u>100</u> GB
Network Adapter	Bridged Adapter

4. NETWORK CONFIGURATION

Commands Executed:

- `sudo systemctl status ssh`
- `sudo apt update`
- `sudo apt install openssh-server -y`
- `sudo systemctl enable ssh`
- `sudo systemctl start ssh`
- `ip a`

Server IP Address: 192.168.137.193

Network Interface: Bridged Network

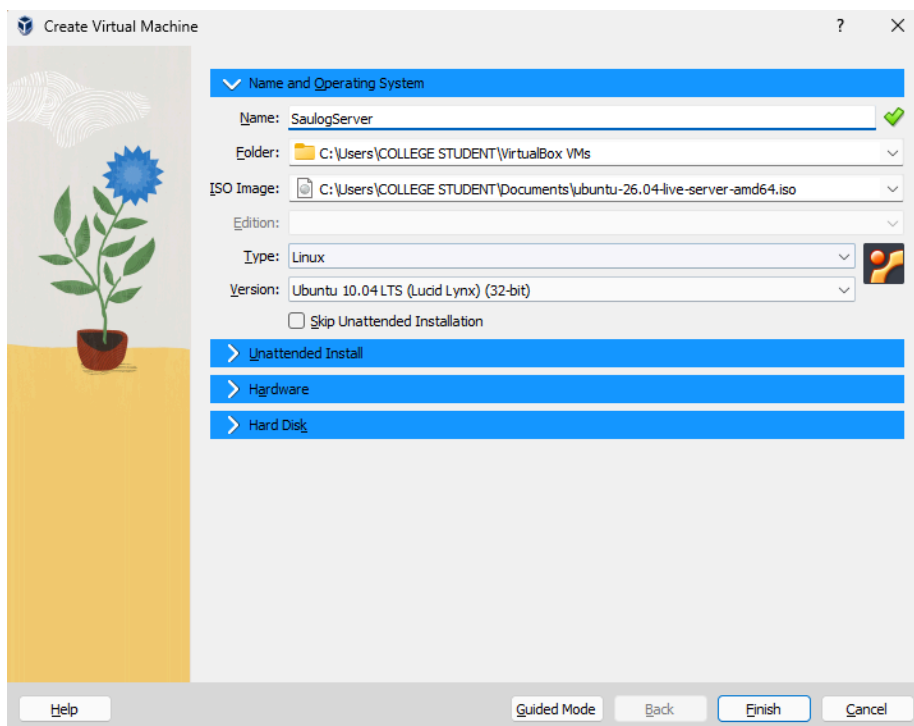
6. REMOTE CONNECTION

Connection Details:

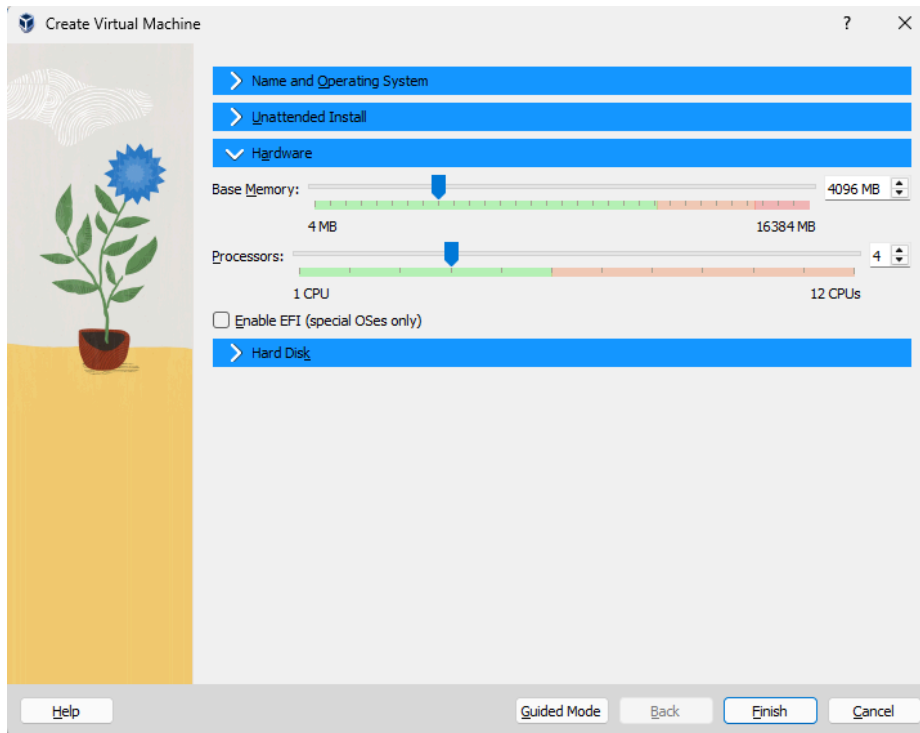
Parameter	Value
Server IP	<u>192.168.137.193</u>
Username	nicsy
SSH Port	ssh nicsy@192.168.137.193

Documentation:

Setting up the Virtual Machine



Set the base memory to 4096 and 4 CPUs, and 100gb hardware



Create Virtual Machine

> Name and Operating System

> Unattended Install

> Hardware

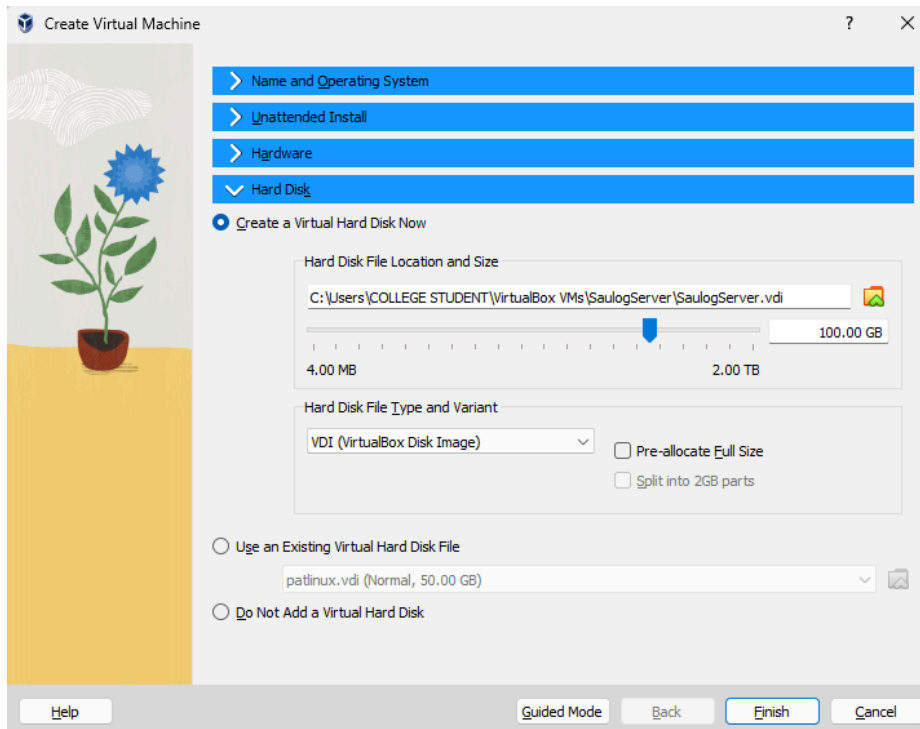
Base Memory: 4096 MB

Processors: 4

☐ Enable EFI (special OSes only)

> Hard Disk

Help Guided Mode Back Finish Cancel



Create Virtual Machine

> Name and Operating System

> Unattended Install

> Hardware

> Hard Disk

☒ Create a Virtual Hard Disk Now

Hard Disk File Location and Size

C:\Users\COLLEGE STUDENT\VirtualBox VMs\SaulogServer\SaulogServer.vdi

100.00 GB

Hard Disk File Type and Variant

VDI (VirtualBox Disk Image)

☐ Pre-allocate Full Size

☐ Split into 2GB parts

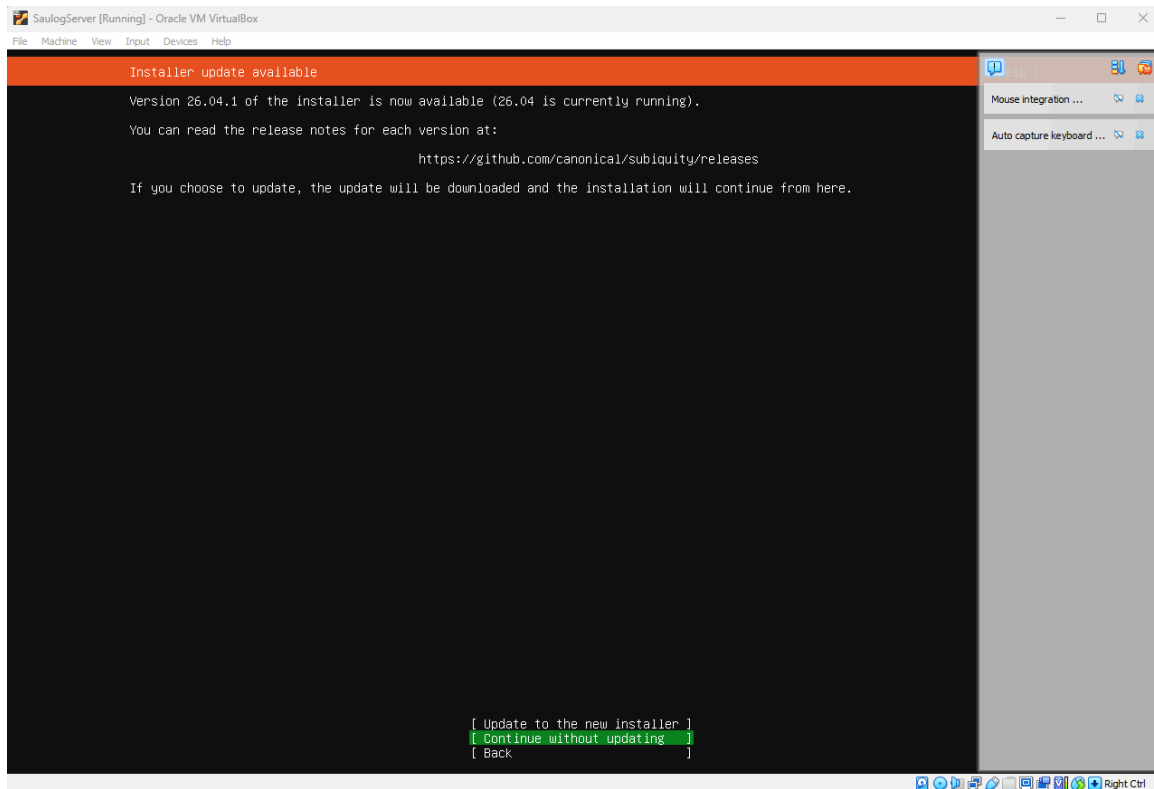
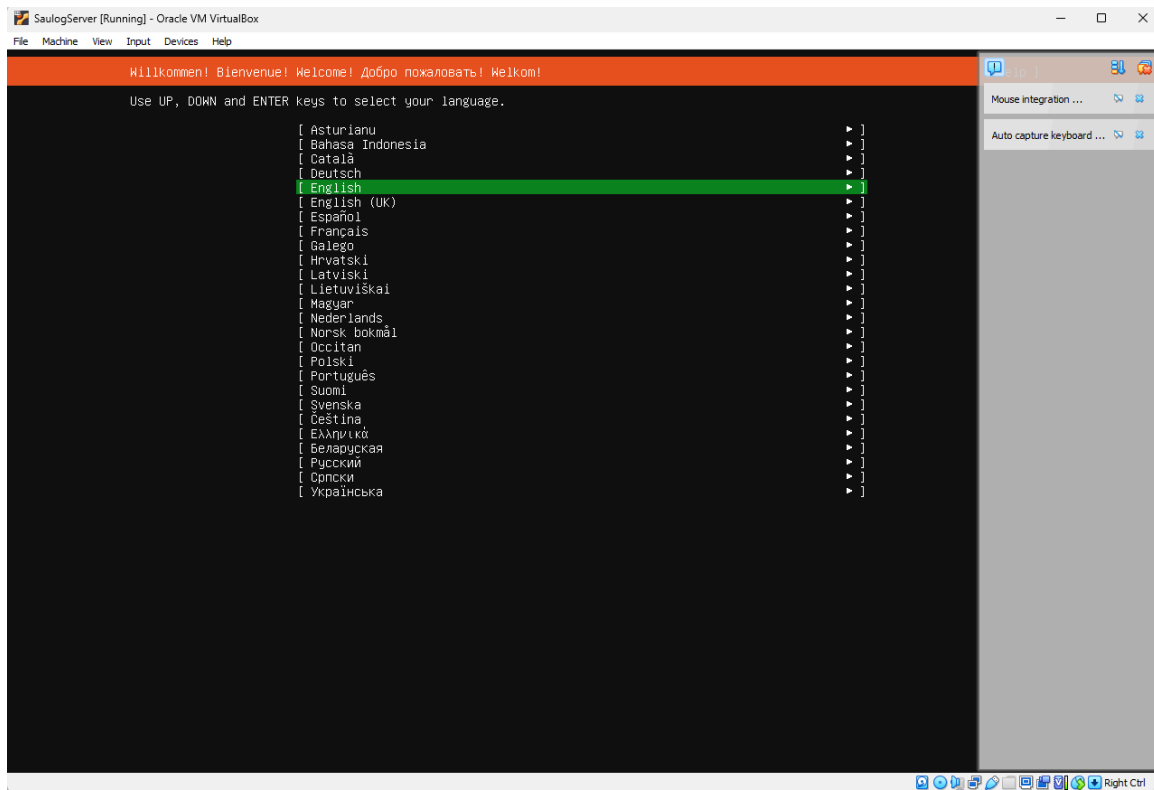
☐ Use an Existing Virtual Hard Disk File

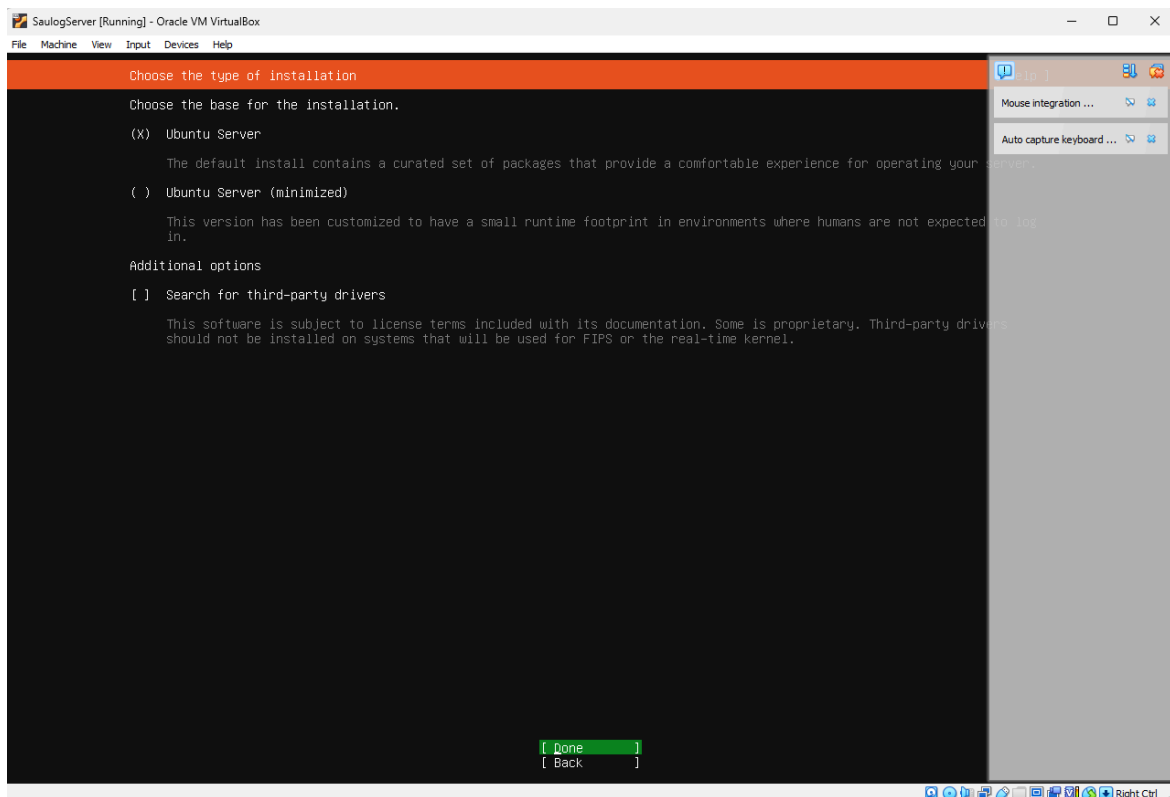
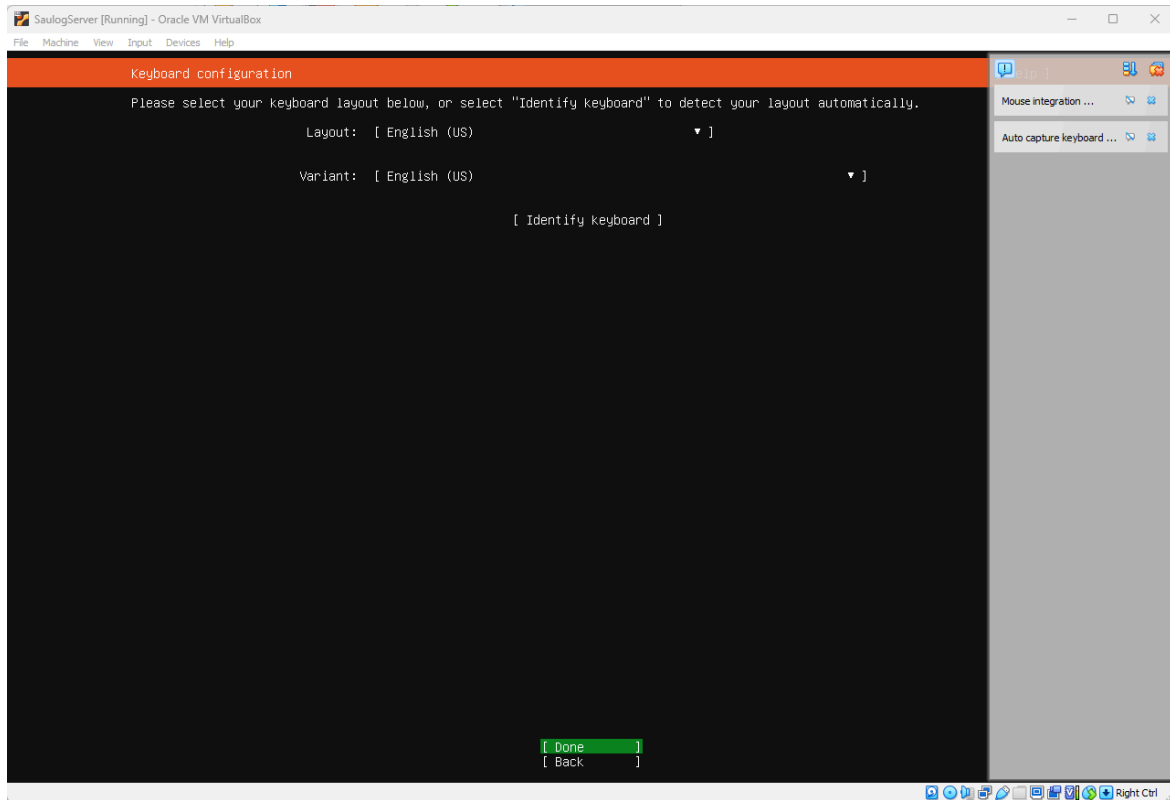
patlinux.vdi (Normal, 50.00 GB)

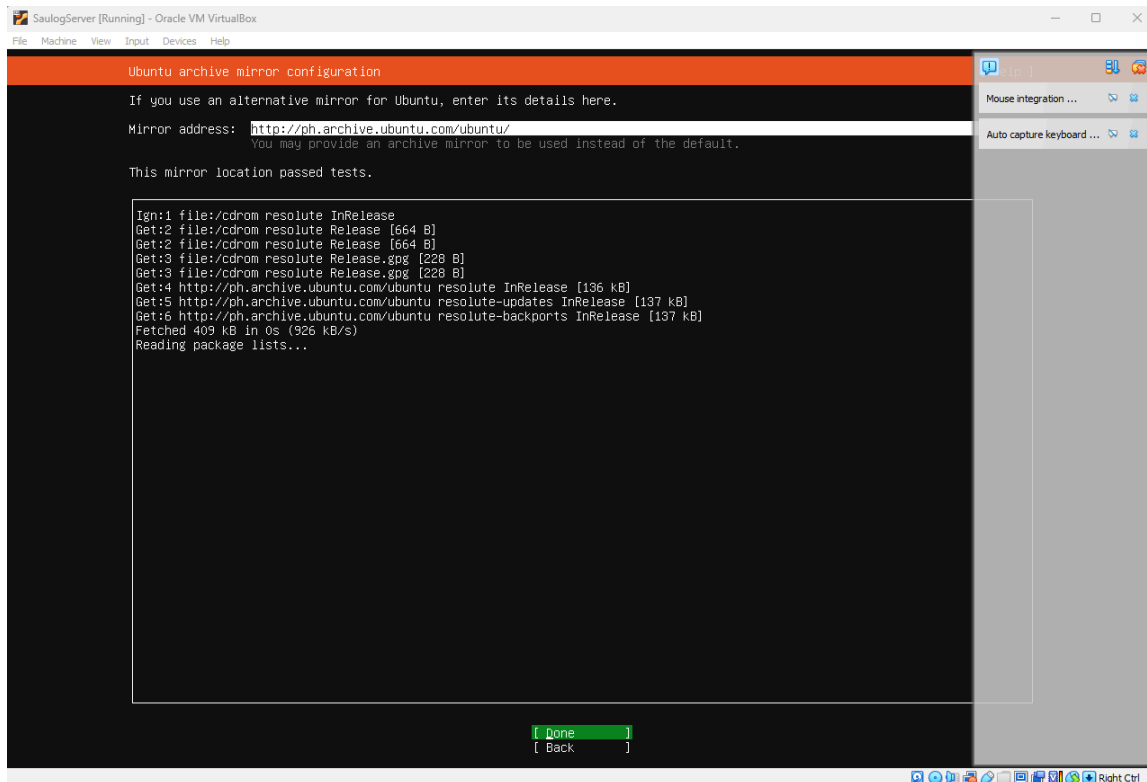
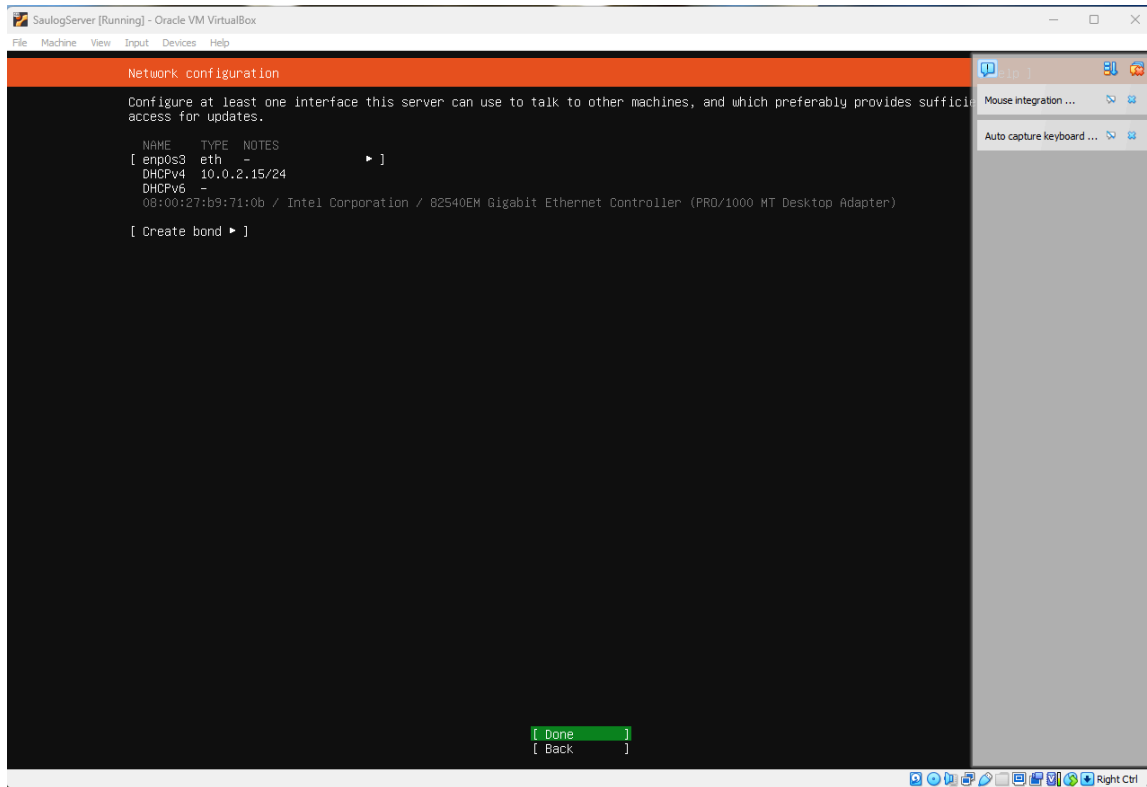
☐ Do Not Add a Virtual Hard Disk

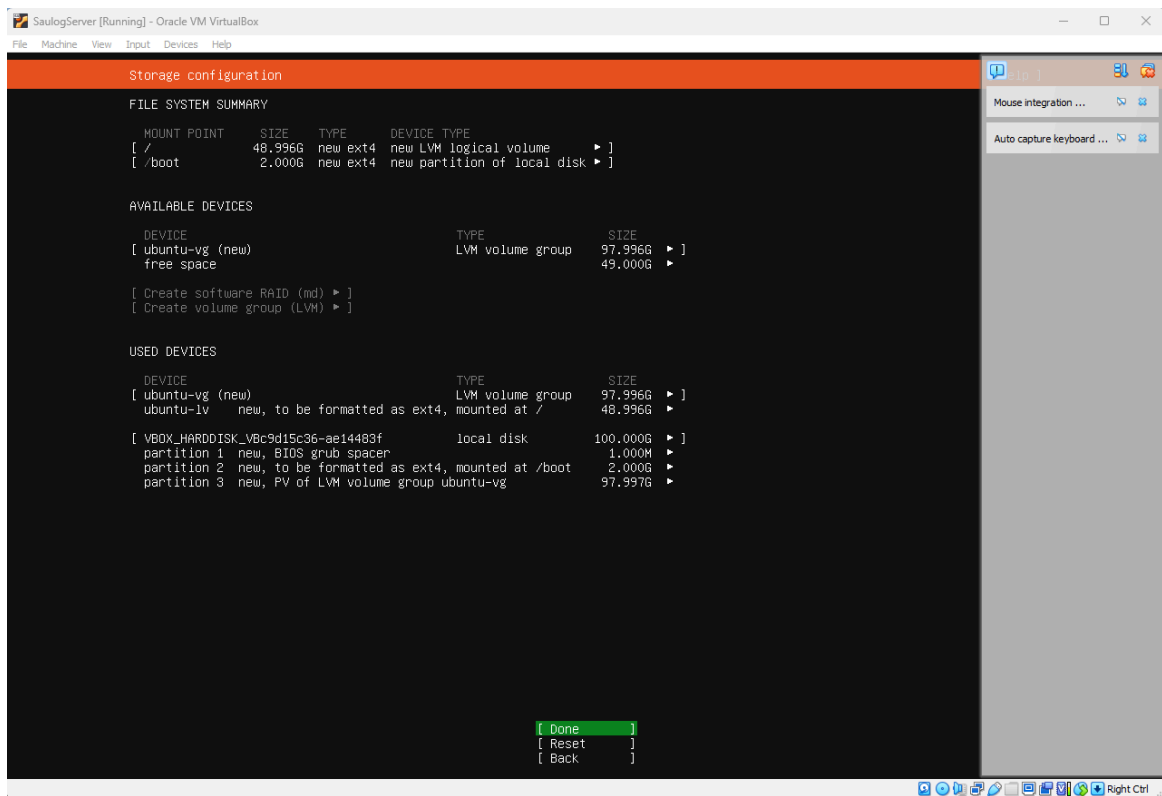
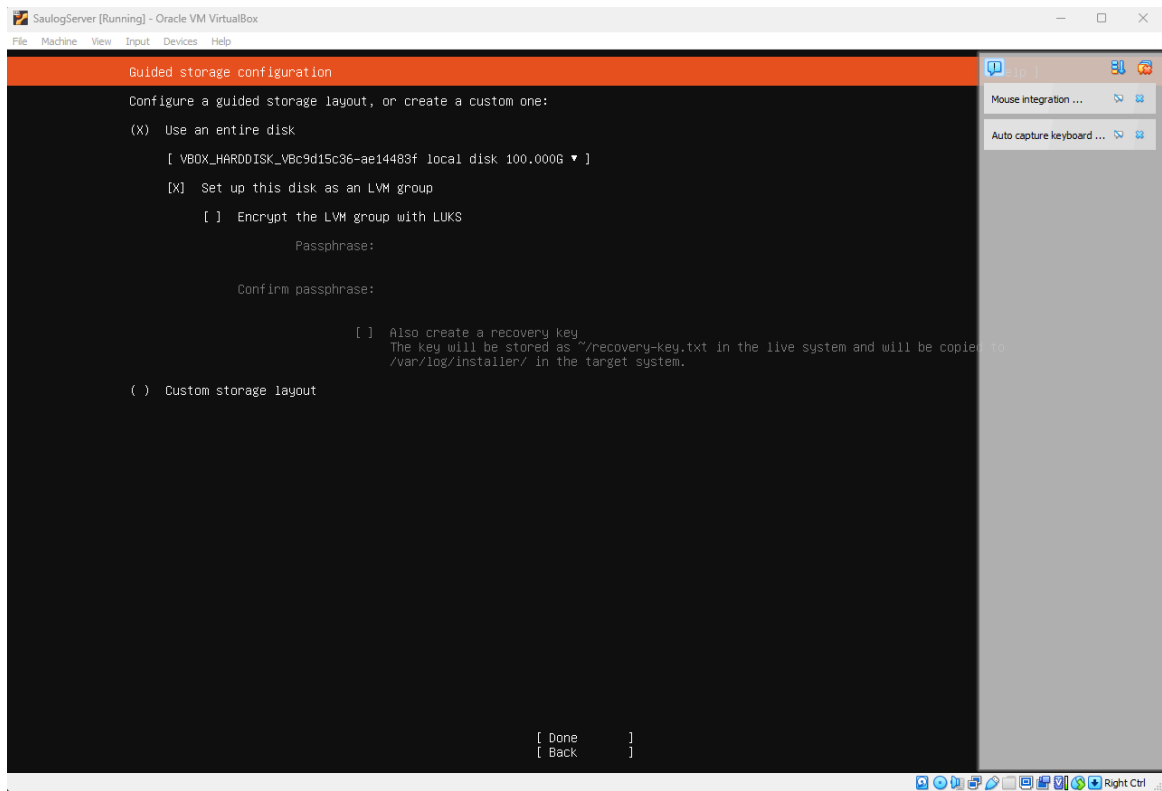
Help Guided Mode Back Finish Cancel

From here, I just read most of it, set it up and move forward, pressing done multiple times until I need to make my account/profile.

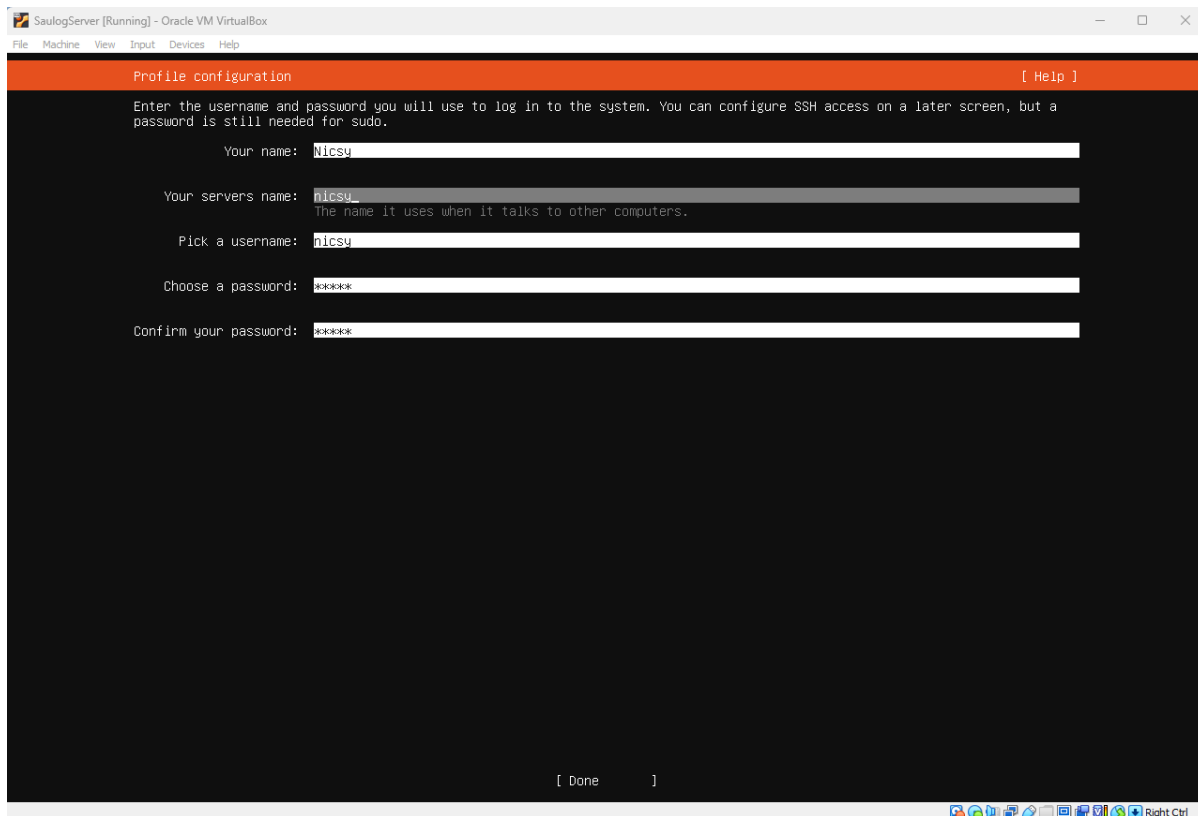




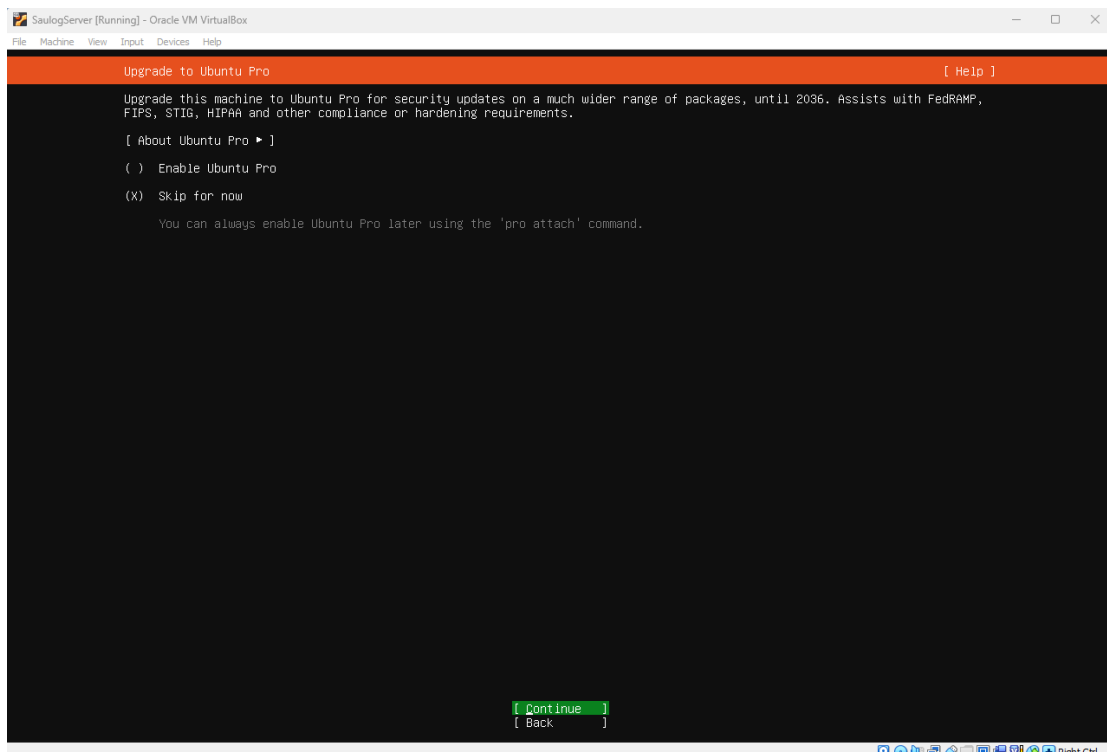


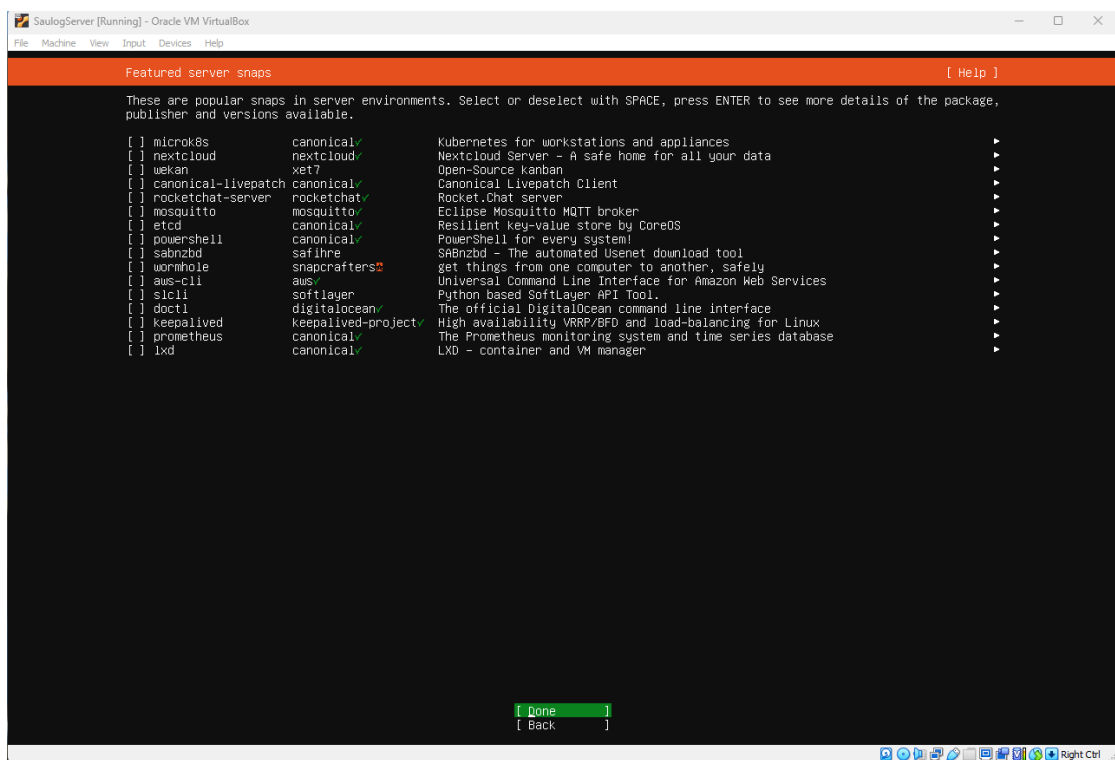
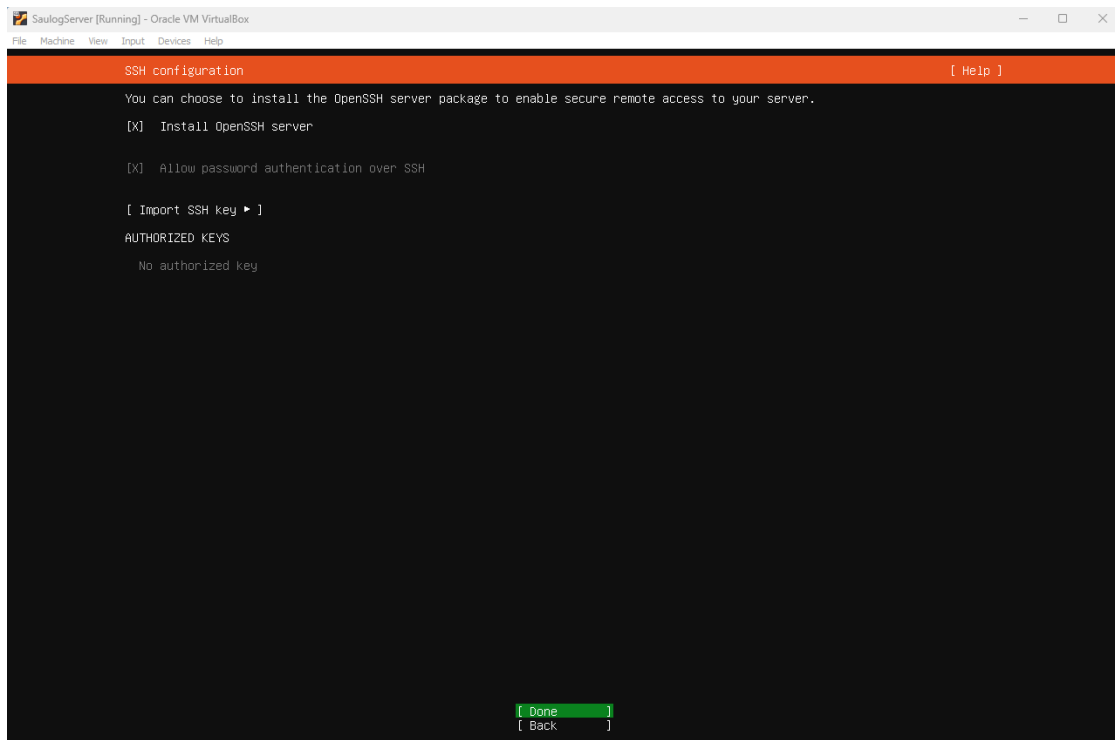


I set up my profile, including server name, username and password.



After configuring my profile, I continue every process that needs to be done.





```
SaulogServer [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Installing system [ Help ]

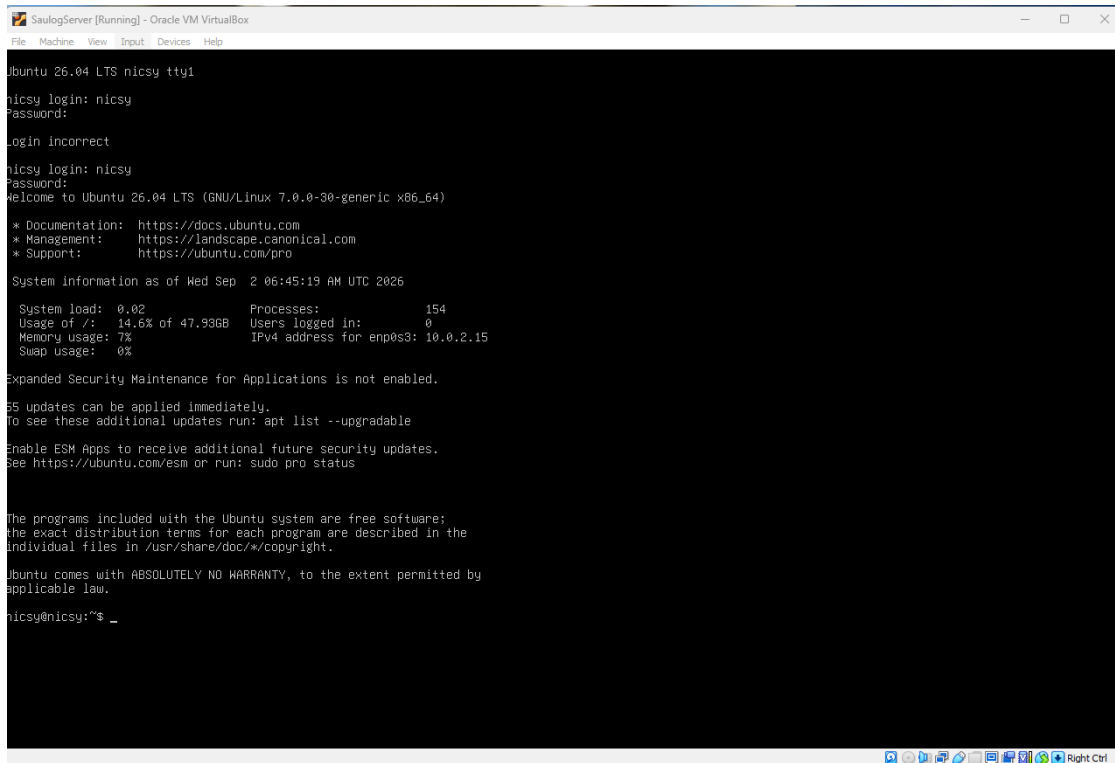
[ 0.854379] configuring partition: partition-2
[ 0.854379] configuring lvm.volgroup: lvm.volgroup-0
[ 0.854379] configuring lvm.partition: lvm.partition-0
[ 0.854379] configuring format: format-1
[ 0.854379] configuring mount: mount-1
[ 0.854379] configuring mount: mount-0
[ 0.854379] executing curtin install extract step
[ 0.854379] curtin command install
[ 0.854379] writing install sources to disk
[ 0.854379] running 'curtin extract'
[ 0.854379] curtin command extract
[ 0.854379] acquiring and extracting image from cp:///tmp/tmpb0903513/mount
[ 0.854379] configuring keyboard
[ 0.854379] curtin command in-target
[ 0.854379] executing curtin install curthooks step
[ 0.854379] curtin command install
[ 0.854379] configuring installed system
[ 0.854379] running 'curtin curthooks'
[ 0.854379] curtin command curthooks
[ 0.854379] configuring apt configuring apt
[ 0.854379] installing missing packages
[ 0.854379] configuring lscs service
[ 0.854379] configuring raid (mdadm) service
[ 0.854379] configuring NWE over TCP
[ 0.854379] installing kernel
[ 0.854379] setting up swap
[ 0.854379] apply networking config
[ 0.854379] writing etc/fstab
[ 0.854379] configuring multipath
[ 0.854379] updating packages on target system
[ 0.854379] configuring pollinate user-agent on target
[ 0.854379] configuring kernel crash dumps settings
[ 0.854379] final kernel configuration
[ 0.854379] configuring target system bootloader
[ 0.854379] installing grub to target devices
[ 0.854379] copying metadata from /cdrom
[ 0.854379] final system configuration
[ 0.854379] calculating extra packages to install
[ 0.854379] installing openssh-server
[ 0.854379] retrieving openssh-server
[ 0.854379] curtin command system-install
[ 0.854379] unpacking openssh-server

[ View full log ]
```

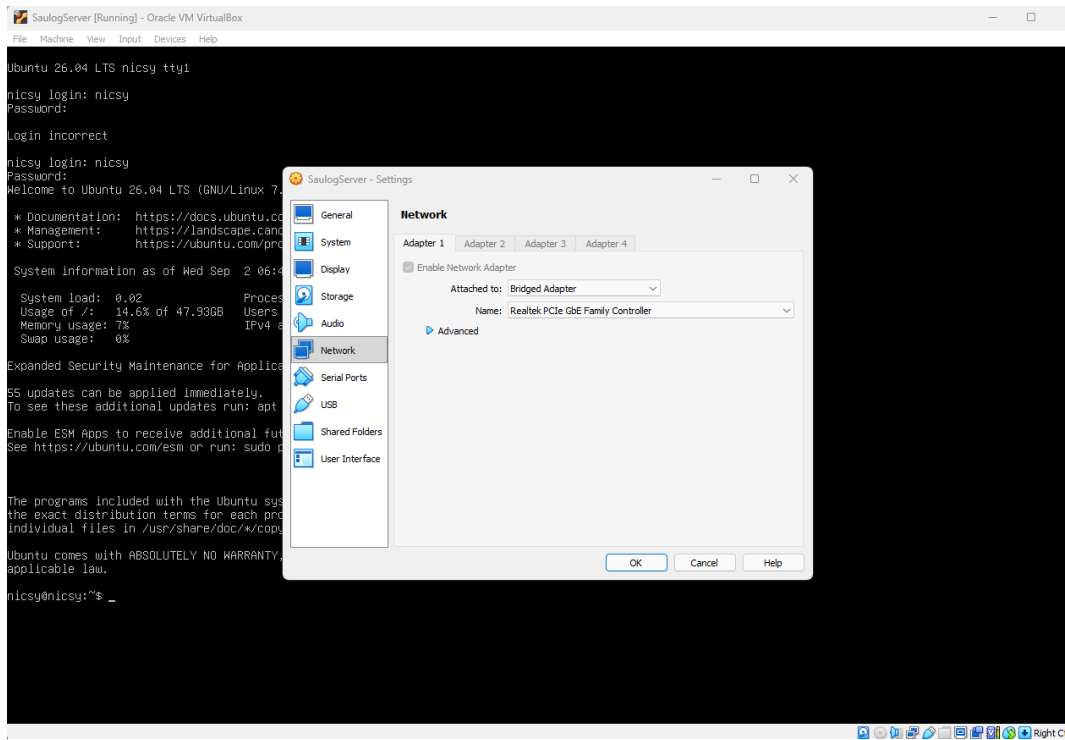
```
SaulogServer [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

[ 0.852192] cloud-init[1052]: The key's randomart image is:
[ 0.854379] cloud-init[1052]: +---[RSA 3072]-----+
[ 0.854535] cloud-init[1052]: | .....oo.. |
[ 0.854577] cloud-init[1052]: | Eo .o o.*.. |
[ 0.854609] cloud-init[1052]: | o + . o o o |
[ 0.854648] cloud-init[1052]: | o o o *++ |
[ 0.854684] cloud-init[1052]: | S +.oo+ |
[ 0.854717] cloud-init[1052]: | oo=0++ |
[ 0.854758] cloud-init[1052]: | .o.*++ |
[ 0.854833] cloud-init[1052]: | .++ |
[ 0.854860] cloud-init[1052]: | +o |
[ 0.854887] cloud-init[1052]: +---[SHA256]-----+
[ 0.854924] cloud-init[1052]: Generating public/private ecdsa key pair.
[ 0.854963] cloud-init[1052]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 0.854998] cloud-init[1052]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 0.855052] cloud-init[1052]: The key fingerprint is:
[ 0.855233] cloud-init[1052]: SHA256:9QZSLKqPSN0qDrtKFcm50qrJaqRMq+16n7gIYV01IQ root@nicsy
[ 0.855321] cloud-init[1052]: The key's randomart image is:
[ 0.855378] cloud-init[1052]: +---[ECDSA 256]-----+
[ 0.855423] cloud-init[1052]: | =. . . |
[ 0.855455] cloud-init[1052]: | E. . . |
[ 0.855541] cloud-init[1052]: | . . .o |
[ 0.855590] cloud-init[1052]: | o.o .o o |
[ 0.855640] cloud-init[1052]: | =. . S o |
[ 0.855709] cloud-init[1052]: | o+= = |
[ 0.855770] cloud-init[1052]: | =o=B o |
[ 0.855827] cloud-init[1052]: | @=+o. |
[ 0.855860] cloud-init[1052]: | #o |
[ 0.855931] cloud-init[1052]: +---[SHA256]-----+
[ 0.855989] cloud-init[1052]: Generating public/private ed25519 key pair.
[ 0.856050] cloud-init[1052]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 0.856183] cloud-init[1052]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 0.856244] cloud-init[1052]: The key fingerprint is:
[ 0.856293] cloud-init[1052]: SHA256:NqJt20UAtaeDHwNL2VRgBRzc9Fq04722IY1NsMy2PK0 root@nicsy
[ 0.856346] cloud-init[1052]: The key's randomart image is:
[ 0.856395] cloud-init[1052]: +---[Ed25519 256]-----+
[ 0.856422] cloud-init[1052]: | .o-B+ . |
[ 0.856477] cloud-init[1052]: | + oo..o . |
[ 0.856525] cloud-init[1052]: | + + o = |
[ 0.856566] cloud-init[1052]: | .B X + = o |
[ 0.856611] cloud-init[1052]: | o & S + o o |
[ 0.856677] cloud-init[1052]: | * @ . . = |
[ 0.856725] cloud-init[1052]: | o B o + |
[ 0.856767] cloud-init[1052]: | o |
[ 0.856828] cloud-init[1052]: | E |
[ 0.857116] cloud-init[1052]: +---[SHA256]-----+

Ubuntu 26.04 LTS nicsy tty1
nicsy login:
```



I changed the network from NAT to Bridged Adapter.



After that, I executed all the commands needed, then looked for my IP Address in order for me to connect my desktop.

```
SaulogServer [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Wed Sep 2 07:56:58 AM UTC 2026

System load: 0.7 Processes: 157
Usage of /: 14.7% of 47.93GB Users logged in: 0
Memory usage: 7% IPv4 address for enp0s8: 192.168.137.193
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

55 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

nicsy@nicsy:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
nicsy@nicsy:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
nicsy@nicsy:~$ sudo systemctl start
Too few arguments.
nicsy@nicsy:~$ sudo systemctl start ssh
nicsy@nicsy:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:9f:d7:ef brd ff:ff:ff:ff:ff:ff
    altname enx0800279fd7ef
    inet 192.168.137.193/24 metric 1024 brd 192.168.137.255 scope global dynamic enp0s8
        valid_lft 604751sec preferred_lft 604751sec
    inet6 fe80::a00:27ff:fe9f:d7ef/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
nicsy@nicsy:~$
```

Then, I opened my desktop, typed ip a, then ssh nicsy@192.168.137.193. It finally connected my server to my desktop.

